Temporal and spatial evidence limits of trained PING

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Abstract

A frozen pyramidal-interneuron gamma (PING) network, whose trained weights remain fixed during evaluation, is tested under complementary temporal and spatial reductions of Modified National Institute of Standards and Technology (MNIST) digit evidence. It classifies continuously streamed digits without retraining, but a duration × input-rate sweep reveals a failure floor below 15 ms. At fixed 200 ms presentation and readout windows, performance remains at 10-class chance through 0.5 Hz and becomes clearly informative by 2 Hz. Separately, foreground pixels are permanently removed from binarized images and presented to both PING and a width-matched artificial neural network (ANN). PING is competitive at intermediate deletion but reaches chance by retention q = 0.02. Together, the curves delimit temporal, event-rate, and spatial evidence regimes for a future variable-rate training experiment.

Methods

Streaming duration and encoding rate

The trained baseline from the canonical PING experiment contains 1024 excitatory (E) and 256 inhibitory (I) cells. It was trained on one MNIST digit per 200 ms trial using random seeds 42, 43, 44. Each seed identifies an independent training run with a reproducible random initialization and data order. Everything here is **inference-only** at the trained timestep $\Delta t = 0.1$ ms; the weights are never updated. The two-dimensional sweep averages over all 3 seeds; the single-stream examples use seed 42.

A stream of digits, each shown for τ ms, is classified in one forward pass. The *only* change from training is a sliding readout window:

1. **Encode** each digit as a Poisson spike train over τ ms across 784 input channels, one per pixel. Each channel generates independent random spikes. The stated encoding rate is the expected number of spikes per second for a full-intensity pixel; lower pixel intensities reduce that rate proportionally.
2. **Concatenate** the per-digit trains into one input stream. At segment index k, the Poisson rate switches instantaneously at the boundary

   $$t_k = k\tau. \quad (1)$$

   Here t_k is the boundary time of segment k, k is the segment index, and τ is the duration of each segment.

3. **Run once** through the trained network at the trained Δt, without retraining.
4. **Integrate evidence** in a non-spiking output leaky integrator, one unit per class. A leaky integrator accumulates incoming spikes while gradually discounting older input:

$$v_{\text{out}}(t) = \beta_{\text{out}} v_{\text{out}}(t-1) + \frac{1-\beta_{\text{out}}}{\Delta t} \boldsymbol{s}^E(t-1) W_{\text{out}}. \quad (2)$$

   Here t is the discrete timestep; $v_{\text{out}}(t)$ is the vector of output-unit states; β_{out} is their leak factor, the fraction of the previous output state retained for one timestep; Δt is the simulation timestep; $s^E(t-1)$ is the E-cell spike vector at the preceding timestep; and W_{out} is the trained E-to-output weight matrix.
5. **Read a sliding window.** Average v_{out} over the trailing τ-window and apply a softmax:

$$\text{logits}(t) = \frac{\Delta t}{\tau} \sum_{u=t-w+1}^{t} v_{\text{out}}(u). \quad (3)$$

$$p(\text{class}, t) = \text{softmax}(\text{logits}(t)). \quad (4)$$

   Here logits(t) is the class-evidence vector; u indexes timesteps in the window; τ is the presentation duration; w is its number of timesteps; and p(class,t) is the softmax-normalized class-probability vector. Softmax converts the logits into non-negative class probabilities that sum to one. The readout-window duration is matched exactly to the current digit's presentation duration:

$$T_{\text{readout}} = T_{\text{presentation}} = \tau. \quad (5)$$

   Here T_{readout} is the duration over which output evidence is averaged, $T_{\text{presentation}}$ is the time for which the digit is shown, and τ is that common duration.

   The corresponding window length is

$$w = \frac{\tau}{\Delta t}. \quad (6)$$

   Every digit is therefore read over exactly its own presentation duration; readout duration is not varied independently. At training the average ran over the *whole* trial; the trailing matched-duration window is the single change.
6. **Predict per segment** at the end of the digit's τ-window according to

$$\hat{c}(t) = \arg\max_{c} p(\text{class} = c, t). \quad (7)$$

   Here c indexes the 10 digit classes, and arg max selects the class with the largest probability.

The output leak is

$$\beta_{\text{out}} = \exp(-\Delta t / \tau_{\text{out}}). \quad (8)$$

Here τ_{out} is the output-unit time constant, which controls how quickly accumulated output evidence decays, and exp denotes the exponential function. The probability trace p(class,t) is the network's online class confidence.

The grid uses presentation durations 10 ms, 15 ms, 25 ms, 40 ms, 50 ms, 75 ms, 100 ms, 200 ms and input rates 5 Hz, 10 Hz, 25 Hz, 50 Hz, 100 Hz, 200 Hz per channel. This gives 48 cells with 1200 classified segments per cell.

To resolve the encoding-rate floor below the grid, additional evaluations use rates 0.01 Hz, 0.025 Hz, 0.05 Hz, 0.1 Hz, 0.25 Hz, 0.5 Hz, 1 Hz, 2 Hz, 3 Hz while holding both presentation and readout at 200 ms. Each cell contains 10 streams of 10 digits for every trained seed. The 5 Hz, 10 Hz, 25 Hz, 50 Hz, 100 Hz, 200 Hz points use the same fixed-duration protocol and come from the corresponding grid row.

Foreground-retention calibration

The spatial protocol uses the same MNIST split and has two parts:

1. Train 3 seeds of a width-matched artificial neural network (ANN) with 784 inputs, one rectified-linear hidden layer of 1024 units, and 10 outputs. The hidden width matches the PING E population, not its recurrent E/I architecture. A rectified-linear unit outputs zero for a negative input and otherwise passes the input unchanged. Training uses 15 epochs, or complete passes through the training set, batches of 256 images per weight update, and learning rate 0.001, the step size of each update.
2. Binarize each held-out image, meaning an image excluded from training, at intensity 0: pixels above the threshold become unit-valued foreground and all others become zero-valued background. Retain every foreground pixel independently with probability q. Retention q = 1 leaves the foreground intact and q = 0 removes it. The ANN calibration uses 10 independent mask realizations per image. The matched comparison uses 100 fixed held-out examples and identical masks for every ANN and PING seed; PING encodes them at 25 Hz for 200 ms.

Results

Streaming classification and temporal evidence

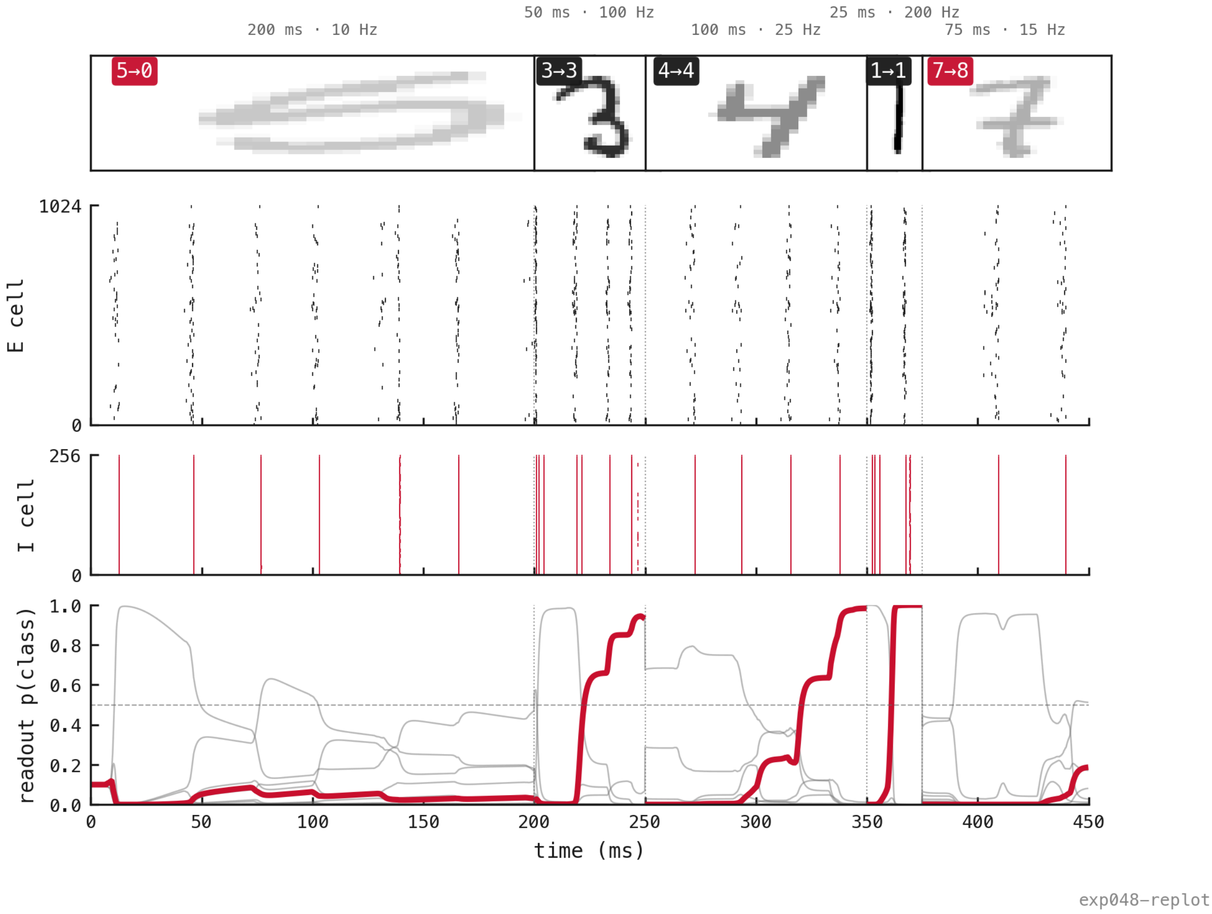

Figure 1: Classification when presentation duration and encoding rate vary between segments. The segment conditions are 200 ms at 10 Hz; 50 ms at 100 Hz; 100 ms at 25 Hz; 25 ms at 200 Hz; 75 ms at 15 Hz. Thumbnail opacity increases with encoding rate. The middle panels plot E- and I-cell spike rasters against time (ms); the lower panel plots class probability against time (ms), with the true class emphasized in red. The label-to-prediction pairs are 5→0, 3→3, 4→4, 1→1, 7→8, giving 3 of 5 correct segments.

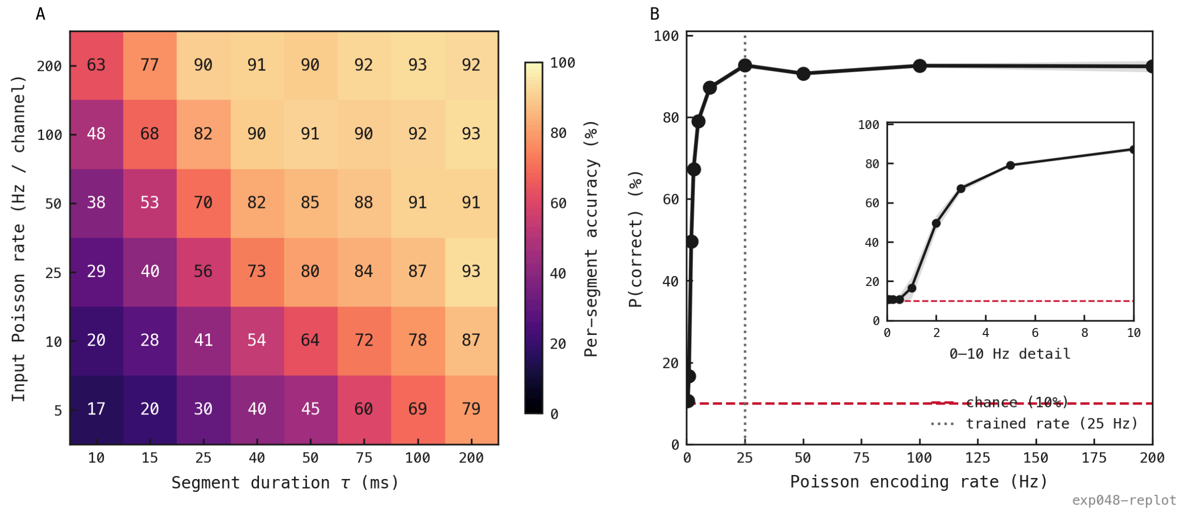

Figure 2: Temporal and encoding-rate limits of the frozen PING classifier. **(A)** Per-segment accuracy (%) is shown for presentation duration (ms, horizontal) and Poisson encoding rate (Hz per channel, vertical), using 1200 segments per cell. **(B)** Probability of a correct classification (%) is plotted against encoding rate (Hz) with presentation and readout fixed at 200 ms. The inset enlarges the linear 0.01–10 Hz interval without changing the axis scale. The dashed line marks 10-class chance and the dotted line the 25 Hz training rate. Accuracy stays at its empty-input floor, the accuracy obtained when almost no input spikes arrive, through 0.5 Hz, becomes informative by 2 Hz, and reaches 79.1% at 5 Hz.

The fixed-duration rate curve distinguishes a nonviable encoder regime from ordinary classification errors under weak evidence. In the variable-condition stream, the first failed segment received 10 Hz for 200 ms, yet that condition reaches 87.3% across the population. Its error is therefore natural trial-level variation, not evidence that 10 Hz is intrinsically too low. The other failed segment, presented at 15 Hz for 75 ms, is likewise above the empty-input rate floor, although its shorter window supplies less total evidence. Rates below 0.5 Hz are not useful operating points; 2 Hz is the lowest clearly informative tested rate and 5 Hz is a practical lower bound for future sweeps.

Spatial evidence calibration

The architecture-matched ANN remains above chance until foreground retention falls to q = 0.005, fewer than one visible foreground pixel per image on average.

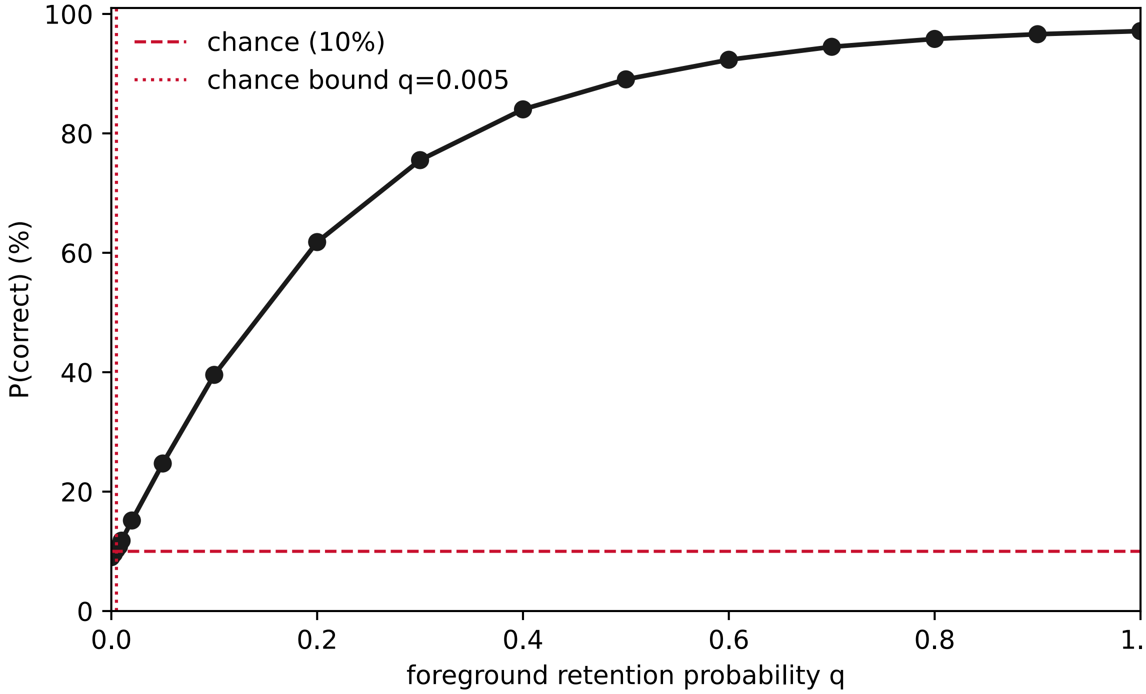

Figure 3: Held-out ANN accuracy as foreground evidence is removed. Probability of a correct classification (%) is plotted against foreground retention q, the independent probability that a foreground pixel remains visible. Points are means across 3 ANN seeds and the band is one standard error, the estimated uncertainty of that mean across seeds. The dashed line marks 10-class chance; the dotted line marks the measured chance-region bound at q = 0.005, the highest tested retention whose 95% confidence interval across seeds still contains chance accuracy.

Under identical masks, neither classifier is uniformly better. With 29.5 visible pixels on average (q = 0.2), PING reaches 68.7% against the ANN's 60.7%. They coincide near 45% at q = 0.1. PING still leads at q = 0.05, then reaches chance at q = 0.02 while the ANN remains above it.

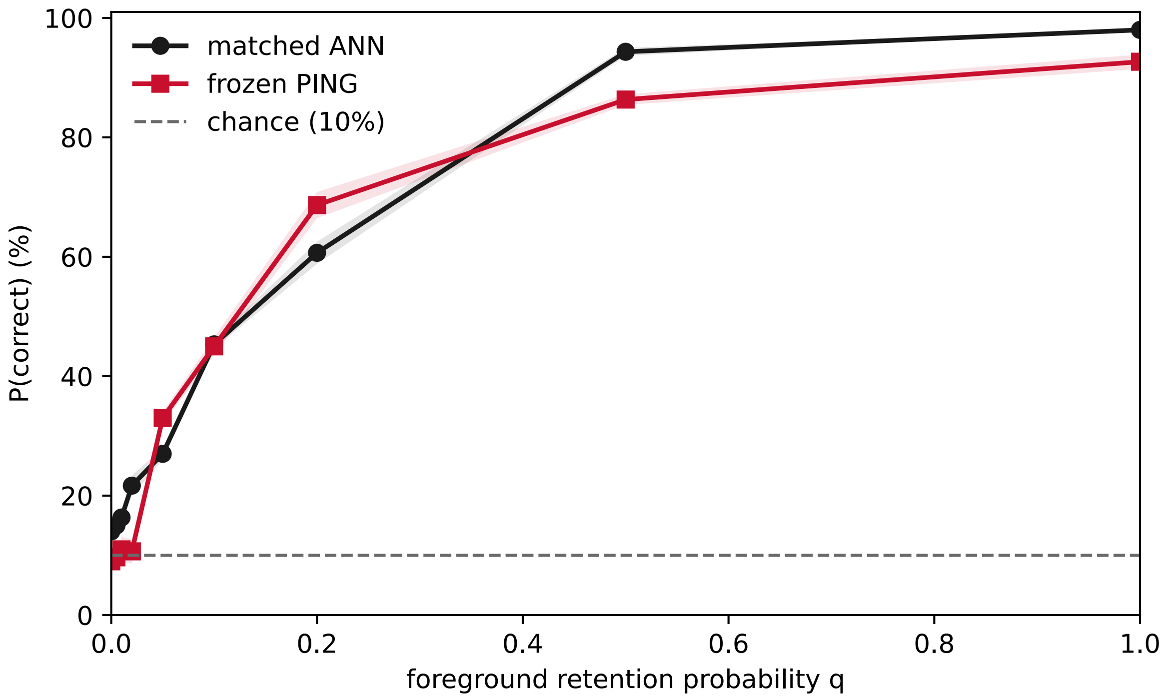

Figure 4: Width-matched ANN and frozen PING accuracy under identical spatial deletion. Probability of a correct classification (%) is plotted against foreground retention q. Black circles denote ANN and red squares PING; bands show one standard error across trained seeds. Each point uses 100 fixed held-out examples and identical independently sampled foreground masks. PING runs at 25 Hz for 200 ms. Neither classifier dominates across the full retention range.

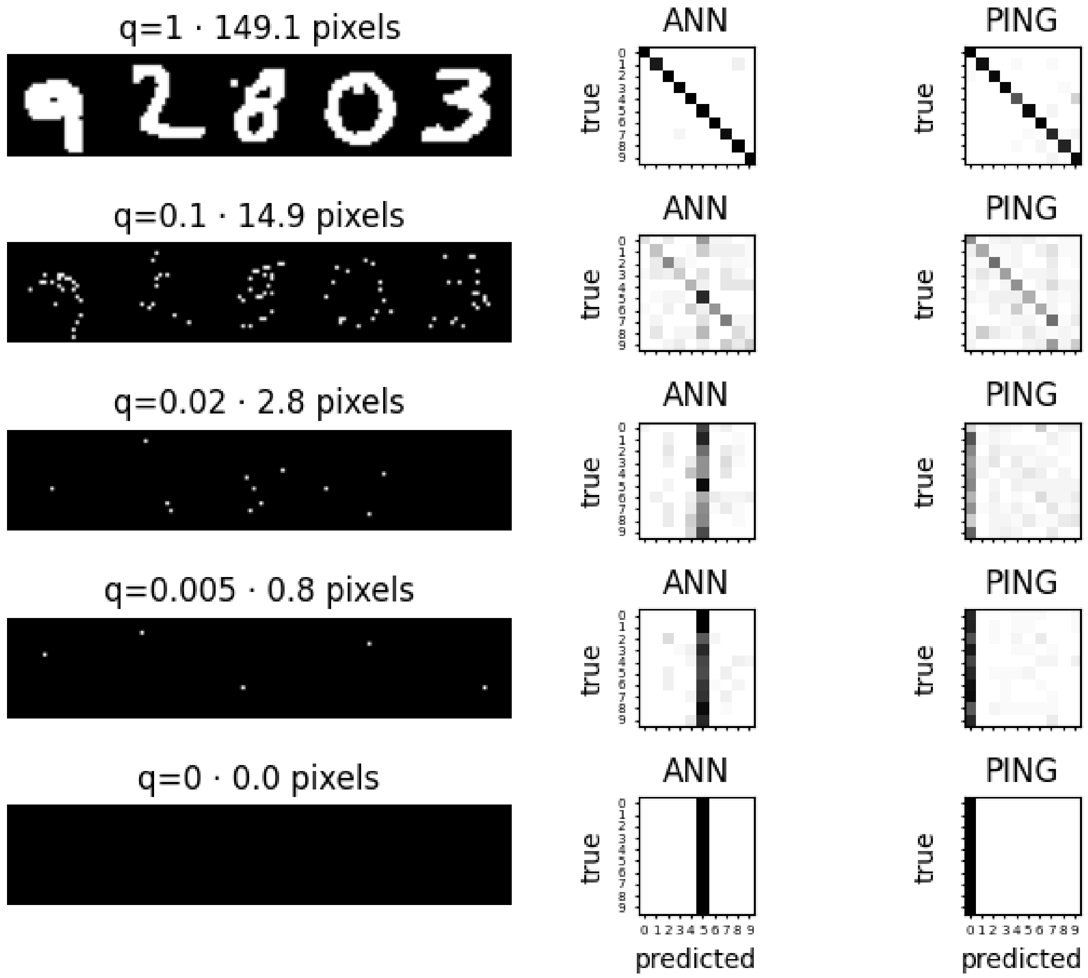

Figure 5: Stimulus and error structure across the matched masking curve. Rows progress from intact input through intermediate masking to the blank control. Left panels show five binarized examples and their mean visible foreground-pixel count. The ANN and PING panels show confusion matrices with true digit on the vertical axis and predicted digit on the horizontal axis; each row is normalized to show the distribution of predictions for one true class. The lowest-retention rows reveal collapse toward model-specific default classes rather than structured digit confusions.

Binarization maps every nonzero antialiased MNIST pixel, including intermediate-intensity pixels introduced to smooth digit edges, to full intensity, so spatial retention is not a second measurement of grayscale contrast. Nevertheless, expected input-event count gives a useful first-order bridge. For an otherwise identical binary image encoded at the masking experiment's 25 Hz ceiling, retaining fraction q gives the same expected event count as retaining the full foreground and using

$$r_{\text{equiv}} = q \cdot 25 \ \ \text{Hz}. \quad (9)$$

Here r_{equiv} is the full-foreground Poisson rate with the same expected event count, meaning the expected total number of input spikes across the image and presentation; q is foreground retention; and 25 Hz is the masking experiment's reference encoding rate.

Thus $q = 0.1$ maps to $r_{equiv} = 2.5$ Hz, within the 2–3 Hz transition of the fixed-duration rate curve. At that retention, PING reaches 45%, compared with 49.7% at 2 Hz in the grayscale rate sweep. This numerical alignment supports including a rate near 2.5 Hz in variable-rate training, but it is not an equivalence of corruptions: spatial masking permanently removes locations, whereas lowering Poisson rate preserves all locations in expectation and changes temporal sampling noise.

Appendix: Proposed filter-matched variable-rate calibration

Prior work motivates the stochastic probe and decoder comparison without answering this calibration's diagnostic question. Wolff and Lindner derive time-dependent moments and autocorrelation for a passive conductance-based membrane driven by filtered Poisson shot noise. They show that transient voltage variability can depart substantially from an effective-time-constant approximation, particularly for larger synaptic events, lower rates, and slower synapses [**1**]. Brigham and Destexhe extend filtered shot-noise analysis to nonstationary conductance input with a continuously varying presynaptic rate, deriving voltage cumulants and showing that a Gaussian approximation can miss distributional skew [**2**]. These results motivate exact finite-window simulation of PING's trained feedforward projection, AMPA conductance, and subthreshold excitatory-cell membrane rather than replacing them with spike counts, one fixed leak factor, or additive Gaussian noise.

Held-out population decoding provides an operational test of what stimulus information a specified response summary and decoder can extract. Quian Quiroga and Panzeri emphasize both the need to separate decoder training from evaluation and the danger of treating near-chance performance from one decoder as proof that the underlying response carries no information [**3**]. Warland et al. directly compare optimized linear and nonlinear neural-network decoders of visual population spike trains, using their agreement to test whether nonlinear extraction recovers information missed by the linear decoder [**4**]. The present proposal applies that logic to PING's frozen input path: independent linear and nonlinear decoders locate a time-averaged-voltage, decoder-relative classification floor as encoding rate falls.

High-level overview

The proposal asks a simple question:

> At a given encoding rate, does the signal delivered to the PING excitatory cells still contain enough information to recognize the digit?

The primary experiment trains a normal artificial neural network (ANN) on a static summary of the same Poisson input that drives PING. The summary is not a raw spike count. Each encoded image passes through the trained input projection, the AMPA synaptic conductance, and a bank of uncoupled, non-spiking excitatory-cell membranes. The membrane voltage is then averaged over the same 200 ms presentation used by PING. This produces one static feature per excitatory cell with the mean and sampling variability created by the actual input filter. Training on a filtered representation follows the shot-noise analyses above, but the exact simulation is retained because low-rate filtered voltage need not be Gaussian [**1**] [**2**].

The probe excludes recurrent excitation, recurrent inhibition, spike threshold, reset, and the trained output layer. It therefore asks whether PING's own feedforward input stage preserves decodable digit evidence before the recurrent circuit acts on it. The new **variable-rate ANN** is trained on these filter-matched features while the encoding rate changes from example to example. Its held-out accuracy curve is the primary calibration used to choose the lower rate for variable-rate PING training. A linear softmax decoder, whose class scores are weighted sums of the probe features, is trained on the same examples as a diagnostic. It tests whether the available digit evidence is already linearly accessible; comparison with the nonlinear decoder follows established population-decoding logic [**3**] [**4**]. The nonlinear variable-rate ANN remains the model used to set the information floor.

This construction cannot use one fixed membrane leak factor. The quoted passive membrane time constant applies only when synaptic conductance is zero. Once input arrives, the total conductance changes both the membrane gain and its effective time constant. The exact conductance-based membrane equation is therefore used to generate ANN features. A local linear approximation is reserved for the complementary transfer-function and Bode analysis.

The proposal has four output groups:

- The **mixed-rate nonlinear psychometric curve** is the primary result. It locates the lowest tested encoding rate at which one decoder trained across the full rate range can classify the time-averaged, filter-matched feedforward representation.
- Three **decoder controls** identify why that primary decoder fails. A mixed-rate linear decoder tests linear accessibility; one mixed-rate coarse temporal linear decoder tests whether averaging over the whole presentation discards useful temporal structure; and separately trained per-rate linear and nonlinear decoders distinguish loss of digit evidence from failure to learn one rate-invariant decision rule.
- The **filter and Bode analysis** is a complementary mechanistic result. It shows how synaptic and membrane filtering shape signal bandwidth, mean, variance, autocovariance, and drive repeatability across encoding rates. Small-signal simulations of the exact conductance probe validate the local transfer approximation before it is interpreted.
- The existing **foreground-retention ANN curve** in Figure 3 is a complementary spatial-evidence result and an implementation sanity check. It does not set the training range.

The interpretation is:

- If the variable-rate ANN and PING both fail, the tested feedforward representation contains too little evidence for either decoder.
- If the variable-rate ANN succeeds but PING fails, the evidence reaches the excitatory-cell input stage, but PING's spiking, recurrent dynamics, readout, or training does not exploit it.
- If both succeed, the rate is already viable.
- If PING succeeds but the variable-rate ANN fails, the probe or ANN is an inadequate surrogate. That outcome invalidates the calibration rather than demonstrating that PING created information.

Within the decoder controls, success of the temporal linear decoder alongside failure of its time-averaged counterpart means that Equation 15 discarded useful temporal structure.

Success of a per-rate decoder alongside failure of the mixed-rate decoder means that digit evidence survives but one rate-invariant readout does not exploit it. Success of the nonlinear decoder alongside failure of the linear decoder means that the retained evidence is not linearly accessible. Failure of all predeclared decoders provides stronger evidence for a floor in this probe-and-decoder family.

The primary result is therefore a **mixed-rate, time-averaged-voltage, decoder-relative classification floor**, not a proof that no conceivable response summary or decoder could recover information at a lower rate.

Filter-matched static signal
For pixel i, the discrete encoder draws

$$S_i(t) \sim \text{Bernoulli}(r\Delta t x_i). \quad (10)$$

Here $S_i(t)$ is the binary input spike at timestep t, r is the maximum encoding rate in spikes per second, Δt is the simulation timestep in seconds, and x_i is the grayscale intensity of pixel i between zero and one.

For excitatory probe cell j, the feedforward AMPA conductance is

$$g_j^{\text{ff}}(t) = \alpha_{\text{AMPA}} g_j^{\text{ff}}(t-1) + \sum_i S_i(t) W_{\text{in}}(i,j). \quad (11)$$

The synaptic decay factor is

$$\alpha_{\text{AMPA}} = \exp\left(-\frac{\Delta t}{\tau_{\text{AMPA}}}\right). \quad (12)$$

Here g^{ff}_j is the feedforward excitatory conductance of probe cell j; $W_{\text{in}}(i,j)$ is the trained weight from pixel i to excitatory cell j; α_{AMPA} is the fraction of AMPA conductance retained for one timestep; and τ_{AMPA} is the AMPA conductance time constant. Equation 11 is the same input-conductance update used by PING, with recurrent contributions omitted.

Each probe cell follows the non-spiking conductance-based membrane equation

$$C_E \frac{dv_j}{dt} = g_{\text{L,E}}(E_L - v_j) + g_j^{\text{ff}}(t)(E_e - v_j). \quad (13)$$

Its instantaneous effective membrane time constant is

$$\tau_{\text{eff}}, j(t) = \frac{C_E}{g_{\text{L,E}} + g_j^{\text{ff}}(t)}. \quad (14)$$

Here v_j is the probe-cell voltage; C_E is the excitatory-cell capacitance; $g_{L,E}$ is its leak conductance; E_L is the leak reversal potential; E_e is the excitatory reversal potential; and $\tau_{\text{eff},j}$ is its conductance-dependent effective time constant. The passive time constant is only the zero-input limit of Equation 14. Input conductance shortens the effective time constant, so a fixed membrane β would be wrong.

The static feature supplied to the variable-rate ANN is

$$z_j = \frac{1}{T} \int_0^T (v_j(t) - E_L) dt. \quad (15)$$

Here z_j is the time-averaged, baseline-subtracted voltage of probe cell j and T is the presentation duration. Subtracting the fixed resting potential only centres the features. The feature is not divided by encoding rate: PING is not told the rate, and removing the rate-dependent change in mean drive would no longer reproduce its input conditions. Equation 15 implements the requested normalization by elapsed time.

Across independent Poisson draws, characterize each feature by

$$\mu_j(r,\boldsymbol{x}) = E\big[z_j \mid r,\boldsymbol{x}\big], \quad \sigma_j^2(r,\boldsymbol{x}) = \mathrm{Var}\big[z_j \mid r,\boldsymbol{x}\big]. \quad (16)$$

A dimensionless drive-to-variability ratio is

$$\mathrm{DVR}_j(r,\boldsymbol{x}) = \frac{\big|\mu_j(r,\boldsymbol{x})\big|}{\sigma_j(r,\boldsymbol{x})}. \quad (17)$$

Here the bold x in Equation 16 denotes the complete grayscale image; μ_j is the mean static feature; σ^2_j is its variance; σ_j is its standard deviation; E denotes an average over repeated Poisson encodings; Var denotes variance over those encodings; and DVR is the drive-to-variability ratio. It measures the mean depolarizing drive of one fixed image relative to variation across its independent Poisson encodings. It is not a class-discriminability signal-to-noise ratio. The decoder psychometric curves provide the population-level test of class information. Report the distribution of these quantities across cells and images rather than hiding their heterogeneity in one grand average.

Because Equation 15 averages a temporally correlated voltage, its variance is not determined by the instantaneous voltage variance alone. Define the conditional voltage autocovariance

$$C_j(t,s \mid r,\boldsymbol{x}) = \mathrm{Cov}\Big[v_{j(t)}, v_{j(s)} \mid r,\boldsymbol{x}\Big]. \quad (18)$$

Then the predicted variance of the averaged feature is

$$\mathrm{Var}\big[z_j \mid r,\boldsymbol{x}\big] = \frac{1}{T^2}\int_0^T \int_0^T C_{j(t,s \mid r,\boldsymbol{x})}\,\mathrm{d}t\,\mathrm{d}s. \quad (19)$$

For the stationary characterization this reduces to

$$\mathrm{Var}\big[z_j \mid r,\boldsymbol{x}\big] = \frac{2}{T^2}\int_0^T (T-u)C_{j(u \mid r,\boldsymbol{x})}\,\mathrm{d}u.$$

Compare both covariance-based predictions with the empirical variance across finite Poisson realizations. This checks the statistics of the exact feature supplied to the decoders rather than only the instantaneous membrane voltage.

The coarse temporal control divides the 200 ms presentation into eight fixed, non-overlapping bins. Its feature for cell j and bin b is

$$z_{j,b}^{\mathrm{bin}} = \frac{8}{T}\int_{(b-1)\frac{T}{8}}^{b\frac{T}{8}} \Big(v_{j(t)} - E_L\Big)\,\mathrm{d}t, \quad b \in \{1,...,8\}. \quad (20)$$

A regularized linear softmax decoder receives the concatenated 1024 × 8 features. This is the only temporal decoder: it is a sensitivity control for information discarded by Equation 15, not a search over bin counts or temporal architectures.

The primary psychometric curve is

$$A_r(r) = P(\text{correct} \mid r, \text{mixed-rate nonlinear decoder on Equation 15}). \quad (21)$$

Chance is estimated empirically. After decoder fitting is complete, repeatedly permute held-out labels relative to the fixed predictions, using the same permutation across rates. For each permutation retain the maximum null accuracy over the evaluated rate grid; let U_{null} be the 95th percentile of that maximum distribution. This gives a simultaneous permutation bound across rates. The primary classification floor is

$$r_{\text{floor}} = \text{lowest} \;\; r \in \mathcal{R} \;\; \text{satisfying} \;\; L_r(r) > U_{\text{null}}. \quad (22)$$

Here A_r is held-out mixed-rate nonlinear-decoder accuracy and P denotes probability. The calligraphic R in Equation 22 is the evaluated rate grid. L_r is the lower confidence bound for A_r. The result r_{floor} is the lowest tested rate whose lower confidence bound exceeds the simultaneous permutation null. Apply the same bound and floor definition to the mixed-rate linear and temporal-linear curves. Per-rate decoders are diagnostic controls and receive their own permutation bounds.

Complementary transfer-function analysis

The exact feature generator uses Equations 10–15. For interpretation only, linearize the membrane around the mean conductance observed for a particular rate, image, and cell. After normalizing to unit gain at zero temporal frequency, the synapse-membrane cascade is

$$H_j(f; r, \boldsymbol{x}) = \frac{1}{(1 + i2\pi f \tau_{\text{AMPA}})(1 + i2\pi f \bar{\tau}_{\text{eff}}, j(r, \boldsymbol{x}))}. \quad (23)$$

Here H_j is the local transfer function; f is temporal modulation frequency in hertz; i is the imaginary unit; and the overbarred effective time constant in Equation 23 is Equation 14 evaluated at the mean feedforward conductance for the selected operating point. Encoding rate is not the horizontal axis of a Bode plot. It selects the operating point and therefore selects one member of a family of transfer curves. This explicit dependence on the temporal filter is consistent with filtered-shot-noise analyses in which synaptic kinetics determine the voltage statistics [**1**] [**2**].

The two local corner frequencies are

$$f_{\text{AMPA}} = \frac{1}{2\pi \tau_{\text{AMPA}}}, \quad f_{\text{mem}}, j = \frac{1}{2\pi \bar{\tau}_{\text{eff}}, j}. \quad (24)$$

Here f_{AMPA} is the AMPA corner frequency and $f_{\text{mem,j}}$ is the local membrane corner frequency. The larger time constant produces the lower corner. Because the membrane time constant changes across rates, images, and cells, plot median Bode magnitude with an interval across those operating points rather than presenting one membrane curve as universal.

Validate this local approximation by applying a 10% sinusoidal modulation to all nonzero pixel rates around each representative operating point, simulating the exact conductance probe after burn-in, and estimating voltage gain and phase over repeated cycles. Compare empirical and predicted gain and phase at every tested frequency. Report median absolute relative gain error and median absolute circular phase error across cells and images. Interpret the local Bode curve only over frequencies where gain error is at most 10% and phase error at most 10 degrees; elsewhere show the failed validation and make no mechanistic claim from the linearized curve.

The finite presentation average in Equation 15 adds the boxcar magnitude

$$B_T(f) = \left|\frac{\sin(\pi f T)}{\pi f T}\right|. \quad (25)$$

Here B_T is the magnitude response of a T-second averaging window, with value one at zero frequency by continuity. Show both the intrinsic cascade in Equation 23 and the task-level response formed by multiplying Equations 23 and 25. This separates cellular filtering from the additional low-pass effect of averaging over the full presentation.

For independent, locally linear filtered Poisson inputs, the stationary moments provide an analytic check:

$$E[y_j] = \sum_i \lambda_i \int_0^\infty h_{\mathrm{ij}}(u)du, \quad \mathrm{Var}[y_j] = \sum_i \lambda_i \int_0^\infty h_{\mathrm{ij}}(u)^2 du. \quad (26)$$

Here y_j is the locally linear filtered signal at cell j; λ_i is the Poisson event rate of pixel channel i; h_{ij} is the impulse response from pixel i to cell j, including its input weight; and u is time. Compare these predicted moments with a long simulation of the exact non-spiking conductance model. Agreement validates the local linear approximation. Disagreement does not invalidate the ANN experiment, which uses the exact simulation, but it limits interpretation of the Bode and analytic signal-to-noise results.

Steps

1. **Lock the roles before implementation.** Treat A_r, the mixed-rate nonlinear curve on Equation 15, as the primary calibration. Treat the mixed-rate linear, coarse temporal linear, and per-rate decoder curves as diagnostic controls. Treat the covariance and validated Bode analyses as complementary mechanistic results. Retain A_q, the existing foreground-retention curve in Figure 3, as a complementary spatial result and sanity check only.
2. **Reuse the established data partitions and baselines.** Use the MNIST training and held-out partitions already used by the foreground-retention experiment. Reuse the three frozen PING baselines and their trained input matrices. Do not use held-out labels while designing the probe, selecting a normalization, or checking the transfer approximation.
3. **Fix the task timing and rate grid.** Use T = 200 ms for every example, matching the presentation and readout window in Figure 2. Evaluate 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, and 25 Hz. These points resolve the current 2–3 Hz PING transition and retain the 25 Hz trained-rate endpoint.
4. **Build the exact feedforward probe.** For each frozen PING seed, route Poisson input spikes through its trained W_{in} and AMPA conductance using Equations 10–12. Apply the same conductance-based excitatory membrane update used by PING, but disable recurrence, threshold, reset, refractory state, adaptation, and the output layer. Vectorize this as 1024 uncoupled probe cells. This is the primary feature generator; no fixed membrane time constant or single membrane β is inserted.

5. **Validate the probe implementation.** At zero input, verify that every probe remains at its resting potential. For controlled constant conductances, compare the numerical voltage update and effective time constant with Equations 13 and 14. For Poisson drive, verify the AMPA conductance trace against Equation 11. These unit checks must pass before any classifier is trained.
6. **Separate stationary filter characterization from the finite task window.** For the mechanistic analysis, drive the non-spiking probe for long enough to discard its initial transient, then estimate stationary conductance and voltage distributions across training images, cells, rates, and repeated Poisson draws. For the ANN dataset, restart the probe from the same initial state used by PING and retain exactly the first 200 ms. The long run estimates steady-state filter statistics; the finite run generates the task-matched ANN feature. Do not substitute one for the other. Transient voltage moments can differ substantially from their stationary limits [**1**] [**2**], so the task-matched feature must include the same startup transient and observation window as PING.
7. **Measure the equivalent static signal and its averaging variance.** Apply Equation 15 to every finite probe trace. Across repeated encodings, estimate the mean, variance, and DVR in Equations 16 and 17, together with the median, interquartile range, skewness, autocovariance, and correlation time of the voltage. Use Equations 18 and 19 to predict the variance of the averaged feature and compare it with empirical finite-window variance. Plot distributions across cells and images against encoding rate. This establishes whether the low-rate regime is dominated by sparse shot-to-shot fluctuations, whether an approximately Gaussian middle regime emerges, and whether relative variability falls as rate increases. Do not assume that absolute variance approaches zero at high rate: filtered conductance-driven voltage distributions can change both scale and shape with drive [**1**] [**2**].
8. **Validate and then interpret the complementary Bode analysis.** At representative low, middle, and high encoding rates, compute the local effective time constants from stationary probe states. Plot Equation 23, mark the corner frequencies from Equation 24, and add the task-level response from Equation 25. Run the exact 10% sinusoidal-modulation validation specified above at the same frequencies and operating points. Plot predicted and empirical gain and phase together, report their errors, and interpret the linearized curve only where both predeclared error tolerances pass. Separately validate the local moment prediction in Equation 26 against exact simulation.
9. **Generate mixed-rate training features.** For every MNIST training presentation, sample one rate uniformly from the Step 3 grid, generate a fresh Poisson encoding, run the exact finite-window probe, and retain both the 1024 values from Equation 15 and the 1024×8 values from Equation 20. Do not give the sampled rate to any decoder as an extra feature. Fresh spike draws on every epoch prevent memorization of particular realizations. Split training images into decoder training and validation subsets before selecting regularization or stopping epochs; held-out test images remain untouched.
10. **Train the mixed-rate decoders.** Use a conventional nonlinear classifier with one rectified-linear hidden layer of 1024 units and 10 outputs. Its input dimension is 1024 because it decodes probe-cell features after W_{in}, whereas the existing foreground-retention ANN receives 784 pixel values. Pair each decoder with one frozen PING input projection

and repeat for seeds 42, 43, 44. On the same Equation 15 features, train a regularized linear softmax decoder. On Equation 20 features, train one regularized coarse temporal linear softmax decoder. Keep both linear curves diagnostic. No decoder shares learned classification weights with PING.

11. **Train per-rate decoder controls.** At each rate in Step 3, generate fresh training and validation realizations at that rate and separately fit linear and nonlinear decoders on Equation 15 features, using the same predeclared model families as Step 10. These controls estimate what a rate-specific readout can extract; they do not set the variable-rate training range and are never given the numerical rate as an input.
12. **Generate all psychometric curves by inference only.** Freeze every decoder. At each rate, evaluate the same fixed held-out images using Poisson draws shared across compatible decoders and seeds, plus additional independent draws to measure sampling variation. Identify the mixed-rate nonlinear curve as A_r. Plot the mixed-rate linear, coarse temporal linear, and per-rate linear and nonlinear curves as diagnostics. No decoder weights or hyperparameters are updated during this sweep.
13. **Fit curves and determine permutation-relative floors.** Fit isotonic regression to every mixed-rate curve and bootstrap held-out images, Poisson draws, and model seeds hierarchically. Generate the simultaneous held-out label-permutation bound specified before Equation 22. Apply Equation 22 to A_r and round the resulting primary floor upward to the next tested rate. Apply the corresponding empirical null to each diagnostic curve. Report gaps between averaged and temporal summaries, mixed-rate and per-rate training, and linear and nonlinear decoders.
14. **Compare the primary curve with frozen PING.** Compare A_r with the fixed-duration PING rate curve in Figure 2 without retraining PING. Report both floors and the gap between them. Interpret that gap as a limitation downstream of the frozen input projection, not as a pure measure of recurrence alone. Use the diagnostic gaps from Step 13 to state whether failures arise from whole-window averaging, the requirement for one rate-invariant readout, linear accessibility, or the broader probe-and-decoder family.
15. **Retain foreground masking as a complementary sanity check.** Reuse A_q from Figure 3 without rerunning its ANN. It was produced by freezing an ANN trained on uncorrupted grayscale images and evaluating independently masked, binarized held-out images. Compare its transition region with A_r and with the event-budget bridge in Equation 9. An accuracy-matched diagnostic rate is

$$r_{\text{info}}(q) = A_r^{-1}\big(A_q(q)\big). \quad (27)$$

Here A_q is the existing foreground-retention ANN curve; $A^{-1}{}_r$ is the inverse of the fitted variable-rate curve; and r_{info} is the lowest rate at which A_r reaches the accuracy observed at retention q. This lowest-rate definition handles flat sections of the fitted curve. Evaluate Equation 27 at $q = 0.1$ and compare it with the 2.5 Hz equal-event prediction. This is an order-of-magnitude check, not a required equality: spatial deletion, temporal undersampling, input representation, and ANN training differ. A gross disagreement should trigger inspection of the encoders, weights, feature scaling, and data pairing; a modest disagreement is scientifically expected.

16. **Choose the variable-rate PING training range.** Use the rounded primary A_r classification floor from Step 13 as the lower endpoint and 25 Hz as the upper endpoint. Sample uniformly from the retained discrete rate grid so every included operating condition receives equal exposure.

TODO

1. [ ] Replace Figure 1 with a representative streaming example. Generate a fixed set of candidate streams under the same conditions, select the stream whose error count is closest to that expected from the measured cell accuracies, and record the candidate set, selection rule, and chosen seed. Prefer one failure, or at most two, without selecting by digit identity.
2. [x] Reuse A_q from Figure 3 without rerunning the foreground-retention ANN.
3. [ ] Implement and unit-test the uncoupled non-spiking feedforward probe in Steps 4 and 5 without changing the production PING engine.
4. [ ] Run the stationary and finite-window probe characterizations in Steps 6 and 7; report feature mean, empirical and autocovariance-predicted averaging variance, DVR, correlation time, and distribution shape against encoding rate.
5. [ ] Produce the local Bode analysis, exact small-signal gain-and-phase validation, and analytic-moment validation in Step 8.
6. [ ] Generate time-averaged and eight-bin filter-matched training features and train the three mixed-rate decoders for seeds 42, 43, 44.
7. [ ] Train the per-rate linear and nonlinear decoder controls in Step 11.
8. [ ] Freeze all decoders and run the shared inference-only rate sweep to generate A_r and every diagnostic curve.
9. [ ] Fit and bootstrap the curves, generate the simultaneous permutation null, and report the primary and diagnostic classification floors.
10. [ ] Compare A_r and its diagnostic controls with the frozen PING curve in Figure 2 and select the lower endpoint for variable-rate PING training.
11. [ ] Compare A_q from Figure 3 with A_r and the Equation 9 event-budget bridge as a complementary sanity check.
12. [ ] After review, run the variable-rate PING training experiment over the selected rate range.

References

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